Lecture 5

Double/Debiased Machine Learning for Treatment and Structural Parameters

ECON6083: Machine Learning in Economics

Today's Roadmap

#	TOPIC	FOCUS
1	Naive ML Problems	Why ML fails for causal inference
2	Neyman Orthogonality	Orthogonal score construction
3	Cross-Fitting	DML1 vs DML2, sample splitting
4	Asymptotic Theory	$N^{-1/4}$ convergence
5	Implementation	DoubleML framework, 401(k) study

Part 1

The Problem with Naive Machine Learning

The Partially Linear Regression (PLR) Model

Structural equations:

$$Y = D\theta_0 + g_0(X) + \zeta, \quad E[\zeta \mid D, X] = 0$$

$$D = m_0(X) + V, \quad E[V \mid X] = 0$$

Parameters:

- θ_0: causal effect (target parameter)
- $g_0(X) = E[Y - D\theta_0 \mid X]$: outcome nuisance function
- $m_0(X) = E[D \mid X]$: propensity score nuisance

Challenge: When $\dim(X)$ is large, g_0 and m_0 cannot be estimated parametrically without severe misspecification

The Naive Plug-in Estimator

Naive approach: Estimate $\hat{g}(X)$ using ML, then solve:

$$\hat{\theta}_{naive} = \left(\frac{1}{n}\sum D_i^2\right)^{-1} \frac{1}{n}\sum D_i(Y_i - \hat{g}(X_i))$$

The problem: ML minimizes prediction error, not parameter bias. It doesn't know θ_0 is special — it just wants $Y \approx D\theta + g(X)$ to fit well.

Result: Catastrophic failure — Regularization bias in $\hat{g}$ contaminates $\hat{\theta}$

Why Naive ML Fails: Regularization Bias

Substituting $Y_i = D_i\theta_0 + g_0(X_i) + \zeta_i$:

$$\hat{\theta}_{naive} - \theta_0 = \underbrace{\left(\frac{1}{n}\sum D_i^2\right)^{-1}\frac{1}{n}\sum D_i\zeta_i}_{\text{Standard noise: } O_p(N^{-1/2})} + \underbrace{\left(\frac{1}{n}\sum D_i^2\right)^{-1}\frac{1}{n}\sum D_i(g_0(X_i) - \hat{g}(X_i))}_{\text{Regularization bias —...}}$$

STANDARD OLS	NAIVE ML
$\hat{\beta} \to \beta_0$ at $\sqrt{N}$-rate	$\hat{g}$ converges at $N^{-1/3}$ or slower
Bias term $\to 0$ fast	Bias term $\approx N^{-1/6}$ — persists!

The Curse of Dimensionality

Optimal convergence rates (Stone 1982; Tsybakov 2009):

	PARAMETRIC (OLS)	NONPARAMETRIC (ML)
Assumption	$g_0(X) = X'\beta$	No functional form assumed
Rate	$N^{-1/2}$	$N^{-2/(4+d)}$ in d dimensions

How dimension affects convergence:

DIMENSION d	RATE	MEETS $N^{-1/4}$ BAR?
$d = 1$	$N^{-0.4}$	Yes
$d = 2$	$N^{-1/3}$	Yes
$d \geq 4$	$N^{-1/4}$ or slower	Borderline or worse

The curse: More features → slower convergence. DML requires $N^{-1/4}$ minimum (Chernozhukov et al. 2018)

The Second Problem: Overfitting Bias

Using same data twice:

- Train ML model on sample: $\hat{g}(X_i)$ fits noise in Y_i
- Evaluate score on same sample: residuals correlate with prediction errors
- Creates "own-observation bias"

Mathematical consequence:

Empirical process terms fail to behave as expected; requires restrictive Donsker class assumptions (often violated by ML)

Two pillars of DML needed:

1. **Orthogonality** — kills regularization bias
2. **Cross-fitting** — kills overfitting bias

The "Double" in DML: Two Nuisance Models

Example: Does college education increase income?

	TREATMENT MODEL $m(X)$	OUTCOME MODEL $g(X)$
Predicts	Who goes to college?	Income ignoring education
Removes	Selection bias	Confounding
Residual	$D - \hat{m}(X)$: "unexpected" enrollment	$Y - \hat{g}(X)$: "unexpected" income

The trick: Regress residuals on each other → causal effect

ML handles high-dimensional X; residuals isolate variation in D

Part 2

Neyman Orthogonality: The First Pillar

From FWL to DML: A Historical Bridge

Frisch-Waugh-Lovell (FWL) Theorem: To estimate θ in $Y = D\theta + X'\beta + \epsilon$:

1. Regress Y on X → residuals $\tilde{Y}$
2. Regress D on X → residuals $\tilde{D}$
3. Regress $\tilde{Y}$ on $\tilde{D}$ → coefficient equals $\hat{\theta}_{OLS}$

	CLASSIC FWL	DML
Model	Linear: $Y = X'\beta + D\theta + \epsilon$	Nonparametric: $Y = g(X) + D\theta + \zeta$
Method	OLS projections	ML (Lasso, RF, NN)
Dimension	$p < N$	$p \gg N$ allowed
Property	Geometric orthogonality	Neyman orthogonality

Same core idea: Partialling out X from both Y and D → isolate clean relationship

Neyman Orthogonality: Definition

Definition: Score ψ is Neyman Orthogonal at (θ_0, η_0) if:

$$\partial_\eta \mathbb{E}[\psi(W; \theta_0, \eta_0)] = 0$$

What this means:

- Expected score is locally flat in the nuisance direction
- Small errors in $\hat{g}, \hat{m}$ don't affect $\hat{\theta}$ to first order
- First-order bias vanishes; only second-order bias remains

Result: DML achieves $\sqrt{N}$-consistent inference despite using ML nuisances

Why Orthogonality Works: The Key Math

Taylor expansion of estimation error:

Let $\hat{g} = g_0 + \delta_g$ and $\hat{m} = m_0 + \delta_m$

$$\hat{\theta} - \theta_0 \approx \underbrace{a_N}_{\text{Noise } O_p(N^{-1/2})} + \underbrace{b_1\delta_g + b_2\delta_m}_{\text{First-order bias}} + \underbrace{c \cdot \delta_g\delta_m}_{\text{Second-order}}$$

	NAIVE	DML
First-order	$b_1 \neq 0$ — bias dominates	$b_1 = b_2 = 0$ — vanishes
Remaining bias	$\delta_g = O(N^{-1/3})$	$\delta_g\delta_m = O(N^{-1/2})$
Result	Biased	$\sqrt{N}$-consistent

Derivation: Naive Score Fails

Naive score: $\psi_{naive} = (Y - g(X) - \theta D) \cdot D$

Take derivative w.r.t. g:

$$\partial_g \mathbb{E}[\psi_{naive}] = -\mathbb{E}[D] \neq 0$$

Result: First-order bias from $\hat{g}$ error enters directly

$$\hat{\theta} - \theta_0 \approx \text{noise} + b \cdot \delta_g$$

where $\delta_g = O(N^{-1/3})$ — bias does not vanish!

Derivation: DML Score Succeeds

DML score: $\psi = (Y - g(X) - \theta D)(D - m(X))$

Step 1: Derivative w.r.t. g

$$\partial_g \mathbb{E}[\psi] = -\mathbb{E}[D - m_0(X)] = 0 \checkmark$$

Step 2: Derivative w.r.t. m

$$\partial_m \mathbb{E}[\psi] = -\mathbb{E}[Y - l_0(X)] = 0 \checkmark$$

Result: Both first-order derivatives vanish!

Remaining bias is second-order: $c \cdot \delta_g \cdot \delta_m = O(N^{-1/2})$ — controllable.

Part 3

Cross-Fitting: The Second Pillar

Cross-Fitting: Breaking the Overfitting Link

The problem: Same data for nuisance estimation AND scoring = own-observation bias

Cross-Fitting = K-fold sample splitting

Step 1: Split data into K folds

Step 2: For each fold k:

- Train nuisance models $\hat{g}, \hat{m}$ on I_k^c (all other folds)
- Predict on I_k: get residuals $\tilde{Y}_i, \tilde{D}_i$
- Models never "saw" I_k data → no overfitting

Step 3: Aggregate across all folds

Key insight: Independence breaks the overfitting link

DML1 vs DML2

DML1 (Average-then-Estimate):

- Solve separate $\check{\theta}_k$ for each fold
- Average: $\hat{\theta}_{DML1} = \frac{1}{K} \sum_{k=1}^{K} \check{\theta}_k$

DML2 (Estimate-then-Average) — Recommended:

- Aggregate scores from all folds, solve one global condition:

$$\hat{\theta}_{DML2} = \frac{\sum_{k,i} \tilde{D}_{ik} \tilde{Y}_{ik}}{\sum_{k,i} \tilde{D}_{ik}^2}$$

FEATURE	DML1	DML2
Estimation	Separate θ per fold	Single θ globally
Stability	Sensitive to fold variance	More robust
Recommended	No	Yes (default)

Choosing the Number of Folds

Practical considerations:

K	PROS	CONS
2	Fast, simple	High variance, wastes data
5	Good balance	Standard choice (default)
10	Low variance	More computation

Recommendation: $K = 5$ or $K = 10$ for most applications

Two pillars of DML:

- Orthogonality kills regularization bias
- Cross-fitting kills overfitting bias

Part 4

Asymptotic Theory: The Central Result

The Central Theorem

Theorem (Chernozhukov et al., 2018):
If the score is Neyman orthogonal, cross-fitting is used, and nuisances converge at $N^{-1/4}$:

$$\sqrt{N}(\hat{\theta}_{DML} - \theta_0) \xrightarrow{d} \mathcal{N}(0, \Sigma)$$

Three error sources in decomposition:

TERM	WHAT IT REPRESENTS	CONTROLLED BY
Oracle noise	Sample avg of scores at true params → CLT	—
First-order bias	From nuisance estimation	**Neyman orthogonality**
Empirical process	From estimated nuisances	**Cross-fitting**

Result: $\sqrt{N}$-consistent, asymptotically normal inference

Comparison of Estimator Properties

FEATURE	NAIVE ML	OLS	DML
Nuisance strategy	Single sample	Parametric	Cross-fitting
Score type	Non-orthogonal	Linear	Neyman orthogonal
Bias source	Regularization	Misspecification	2nd order only
Required rate	$N^{-1/2}$ (fast)	N/A	$N^{-1/4}$ (slow)
Valid in high-dim	No	No	Yes

Asymptotic Variance and Inference

Variance formula:

$$\hat{\Sigma} = \frac{\frac{1}{N}\sum_{k,i}\psi^2(W_i;\hat{\theta},\hat{\eta}_k)}{\left(\frac{1}{N}\sum_{k,i}\partial_\theta\psi(W_i;\hat{\theta},\hat{\eta}_k)\right)^2}$$

95% Confidence Interval:

$$[\hat{\theta} - 1.96\cdot\hat{\text{SE}},\ \hat{\theta} + 1.96\cdot\hat{\text{SE}}]$$

Valid inference without knowing nuisance function forms

Part 5

Application: 401(k) Plans and Wealth

Paper: Chernozhukov et al. (2018)

"Double/Debiased Machine Learning for Treatment and Structural Parameters for Treatment and Structural Parameters"
The Econometrics Journal, 2018

Research question: Do 401(k) retirement plans increase net financial wealth, or merely shift savings from other accounts?

ML: DML with cross-fitting using Lasso, Random Forest, and XGBoost as nuisance learners to control for high-dimensional confounders

Data: $N = 9{,}915$ US households from the 1991 Survey of Income and Program Participation (SIPP)

401(k) Study: Variables and Strategy

VARIABLE	DEFINITION	TYPE
Y	Net financial assets (assets — debt)	Continuous
D	401(k) eligibility	Binary
X	Income, age, education, family size, marital status, IRA, home ownership	Controls

Identification: Eligibility determined by employer, not individual choice → conditional on X, eligibility is "as-if random"

OLS problems: Income-savings relationship highly non-linear; many interactions (Age×Income, Education×Income, etc.)

401(k) Study: Results

ESTIMATOR	ML LEARNER	$\hat{\theta}$	STD ERROR	95% CI
Naive OLS	None	$19,559	$2,121	[15,402; 23,716]
DML	Lasso	$8,974	$1,324	[6,379; 11,569]
DML	Random Forest	$8,909	$1,321	[6,320; 11,498]
DML	XGBoost	$8,597	$1,350	[5,951; 11,243]

Source: Chernozhukov et al. (2018)

401(k) Study: Interpretation

Key findings:

1. **OLS severely overestimates** (\$19,559): Residual confounding from income/job quality; linear controls inadequate
2. **DML estimates are consistent** (\$8,500–\$9,000): Different ML learners give similar results — robust causal effect
3. **401(k) eligibility increases net wealth by ~\$9,000**: Significant at conventional levels; evidence against pure substitution

Sensitivity: All specifications with different learners and fold numbers ($K = 3, 5, 10$) yield estimates in the \$8,000–\$10,000 range

Implementation: The DoubleML Framework

```python
import doubleml as dml
from sklearn.ensemble import RandomForestRegressor

# Prepare DoubleML data object
data = dml.DoubleMLData(
    df, y_col="net_tfa",
    d_cols="e401", x_cols=controls
)

# Specify learners for g(X) and m(X)
ml_g = RandomForestRegressor(n_estimators=500)
ml_m = RandomForestRegressor(n_estimators=500)

# Fit DML2 with 5-fold cross-fitting
model = dml.DoubleMLPLR(data, ml_g, ml_m, n_folds=5)
model.fit()
print(model.summary)
```

Choosing ML Learners

Practical guidelines:

DATA CHARACTERISTICS	RECOMMENDED LEARNER
Linear relationships	Lasso, Ridge
Non-linear, smooth	Neural Network, Kernel methods
Interactions, categorical	Random Forest, XGBoost
Mixed / unknown	Ensemble (Super Learner)

Always tune hyperparameters via cross-validation

Need $o(N^{-1/4})$ convergence rate for nuisances — tune carefully

Simulation Validation

Monte Carlo Illustration (simulated DGP for teaching purposes):

DGP: $Y_i = D_i \cdot 2 + X_{i1} + X_{i2}^2 + \varepsilon_i, \varepsilon_i \sim N(0,1), n \in \{500, 1000, 5000\}$

N	OLS BIAS	NAIVE ML BIAS	DML BIAS	DML COVERAGE
500	0.05	0.42	0.08	88%
1000	0.03	0.38	0.04	92%
5000	0.01	0.35	0.01	95%

Values from author simulation. See `scripts/simulate_dml_bias.py` for replication.

Key findings:

- OLS bias $\to 0$ as N grows (correctly specified)
- Naive ML bias **persists** regardless of N (regularization bias)
- DML bias $\to 0$ and coverage $\to$ nominal level

> ***Pedagogical note:*** *These numbers illustrate the qualitative pattern documented in Chernozhukov et al. (2018) and Farrell (2015). For exact replication, run the simulation script.*

Extension: Nonlinear DML

Partially linear model: $Y = \theta D + g(X) + \zeta$ (linear in D)

Fully nonparametric: $Y = g(D, X) + \zeta$

DML extension (Chernozhukov et al. 2018):

- Target: average derivative $\theta = E[\partial g(D, X)/\partial D]$
- Score: $\psi = \partial_\mu \ell(Y; \mu) \cdot (D - m(X))$ where $\mu = g(D, X)$

Application: Dose-response curves, price elasticity as function of income

Sensitivity Analysis

Question: How strong would an unmeasured confounder need to be to invalidate our results?

Partial R^2 framework (Cinelli & Hazlett 2020):

- $R^2_{Y\sim U|X,D}$: Confounder's explanatory power for Y
- $R^2_{D\sim U|X}$: Confounder's explanatory power for D

Robustness value (RV): Minimum strength of unmeasured confounding needed to change conclusion

Interpretation: If RV = 20%, an unmeasured confounder would need to explain 20% of residual variance in both Y and D to nullify the effect.

Common Pitfalls

1. Forgetting cross-fitting

- Same data for nuisance estimation and inference → overfitting bias
- Always use K-fold cross-fitting

2. Using weak ML learners

- Need $o(N^{-1/4})$ rate for nuisances
- Tune hyperparameters via CV; check nuisance fit quality

3. Applying DML when unconfoundedness fails

- DML requires $Y(1), Y(0) \perp D \mid X$
- Check DAGs and identification strategy first

4. Ignoring DML2 vs DML1 distinction

- DML2 (global moment) more stable — use as default

Extension: Multiple Treatments

Setting: $D = (D_1, D_2, \ldots, D_k)$ — multiple policy levers

Model: $Y = \sum_{j=1}^{k} D_j\theta_{0j} + g_0(X) + \zeta$

DML extension:

1. Estimate $\hat{m}_j(X) = E[D_j|X]$ for each treatment
2. Residualize each treatment: $\tilde{D}_{ij} = D_{ij} - \hat{m}_j(X_i)$
3. Run multivariate regression of $\tilde{Y}$ on $\tilde{D}_1, \ldots, \tilde{D}_k$

Key requirement: Each treatment has its own orthogonal score → joint inference valid

Extension: DML with Weak Instruments

Problem: In IV setting, instrument Z may be weak: $Cov(Z, D|X) \approx 0$

DML-IV score (Chernozhukov et al. 2018):

$$\psi = \big(Z - r(X)\big) \cdot \big(Y - g(X) - \theta(D - m(X))\big)$$

where $r(X) = E[Z|X]$ is an additional nuisance function.

Weak instrument issue: First-stage residual variance is small → $\hat{\theta}$ has high variance

Solution: Report first-stage F-statistic; avoid inference when $F < 10$

Application: Price Elasticity of Demand

Question: How does price affect demand for a product?

Endogeneity: Prices set strategically based on expected demand → $Cov(P, U) \neq 0$

DML solution:

- Y = quantity demanded
- D = price
- Z = wholesale cost / competitor price (instrument)
- X = product characteristics, seasonality, demographics

Result: DML-IV with RF nuisances yields consistent elasticity estimates even with 100+ controls

Key Takeaways

CONCEPT	KEY POINT
Naive ML fails	Regularization bias in nuisances contaminates $\hat{\theta}$
Orthogonality	Residualizing D makes $\hat{\theta}$ insensitive to nuisance errors
Cross-fitting	Train nuisances on one fold, estimate on another
DML2	Solve global moment condition — more stable than DML1
Rate condition	Nuisances need $N^{-1/4}$ — met by Lasso, RF, XGBoost

Further Reading

PAPER	AUTHORS	VENUE
Double/Debiased Machine Learning for Treatment & Structural Parameters	Chernozhukov et al. (2018)	*Econometrics Journal*
Inference on Treatment Effects after Selection among High-Dimensional Controls	Belloni, Chernozhukov & Hansen (2014)	*ReStud*
Towards Optimal Doubly Robust Estimation of Heterogeneous Causal Effects	Kennedy (2020)	*Biometrika*
Quasi-Oracle Estimation of Heterogeneous Treatment Effects	Nie & Wager (2021)	*Biometrika*
Double Machine Learning for Treatment and Causal Parameters	Chernozhukov et al. (2016)	arXiv

Data and Code

This lecture:

- Slides source: `_slides/Lecture05-Double/Debiased Machine Learning-Slides.md`
- Exercises: `exercises/Lecture05-Exercise.ipynb`

Key packages:

- Python: `econml` (DML), `doubleml` (R package)

Datasets:

- 401(k) eligibility (Poterba, Venti & Wise 1995)

Thank You!

Next Lecture: Heterogeneous Treatment Effects

Reading: Wager & Athey (2018), "Estimation and Inference of HTE"